

Erwin Quanten

Staff Software Engineer – Go, Kubernetes, AWS, Distributed Systems

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PROFESSIONAL SUMMARY

Staff Software Engineer with 16+ years of experience building backend platforms across **cloud infrastructure, SaaS, fintech, logistics, manufacturing, and enterprise automation**, specializing in **Go 1.17–1.22, Kubernetes 1.23–1.29, AWS, gRPC, Kafka, PostgreSQL 12–16**, and observability-driven distributed systems, with strong ownership of service modernization, API performance, legacy migration, platform reliability, production debugging, and scalable engineering practices that improve deployment quality, reduce operational risk, and support high-volume business workflows.

SKILLS

- **Languages:** Go 1.17–1.22, Java 8–17, Python 3.8–3.11, TypeScript 4–5, Bash, SQL
- **Go Backend:** Gin 1.8–1.10, Echo 4.x, Chi 5.x, gRPC 1.40–1.60, Protocol Buffers 3.x, REST APIs, workers, schedulers, concurrency patterns, context handling
- **Cloud & DevOps:** AWS, EKS, ECS, Lambda, RDS, S3, SQS, SNS, CloudWatch, IAM, Docker 20–24, Kubernetes 1.23–1.29, Helm 3.x
- **Data & Messaging:** PostgreSQL 12–16, MySQL 8, Redis 6–7, Kafka 2.8–3.6, RabbitMQ 3.8–3.12, DynamoDB
- **Architecture:** Microservices, event-driven systems, API gateways, distributed tracing, domain-driven design, clean architecture, service decomposition
- **Observability:** OpenTelemetry 1.x, Prometheus 2.x, Grafana 9–10, ELK, CloudWatch Logs, structured logging, alerting, SLO dashboards
- **CI/CD & IaC:** GitHub Actions, GitLab CI, Jenkins, Azure DevOps, Terraform 1.2–1.6, Helm, Argo CD
- **Security:** OAuth2, JWT, OIDC, IAM least privilege, secrets management, API rate limiting, dependency scanning

EXPERIENCE

Lightedge — Lead Software Engineer

Remote/ Full time

May2022 – Apr 2026

Houston, TX

- Led backend modernization for a cloud infrastructure and managed services platform by building **Go 1.18–1.22** microservices that improved API throughput by **31%** while reducing average response latency by **24%**.
- Designed high-volume **gRPC** services with Protocol Buffers 3.x and streaming patterns to replace slower REST-only integrations, improving internal service communication reliability by **28%**.
- Migrated legacy Node.js and Java background workers into **Go** using goroutines, worker pools, and context cancellation, reducing compute usage by **22%** during peak workloads.
- Implemented production-grade **Kubernetes 1.24–1.29** deployment patterns on AWS EKS with Helm 3.x, readiness probes, autoscaling, and safer rolling updates for customer-facing APIs.
- Built event-driven workflows with **Kafka 3.x**, SQS, and idempotent Go consumers to handle provisioning events, reducing duplicate processing issues by **35%** across infrastructure automation flows.
- Resolved a critical race condition in a Go billing worker by adding mutex-safe cache access, structured retries, and integration tests, preventing incorrect invoice status updates in production.
- Improved PostgreSQL 14–16 query performance by rewriting slow joins, adding targeted indexes, and tuning connection pooling, reducing expensive report generation time by **41%**.
- Implemented OpenTelemetry 1.x tracing with Prometheus and Grafana dashboards, helping engineering teams detect API bottlenecks earlier and reduce mean time to recovery by **30%**.
- Created secure OAuth2 and JWT middleware in **Go** with tenant-aware authorization rules, improving access control consistency across multiple internal and customer-facing services.
- Introduced CI/CD quality gates with GitHub Actions, Docker image scanning, Go test coverage checks, and automated Kubernetes deployments, reducing failed production releases by **26%**.
- Built migration tooling to move selected services from VM-based deployment to containerized EKS workloads, including config separation, secret rotation, health checks, and rollback strategy.
- Mentored backend engineers on **Go concurrency**, error handling, interface design, testing strategy, and production debugging so teams could build simpler services with fewer runtime defects.

Syscom Digital — Staff Software Engineer

Remote/ Full time

Nov 2017 – Apr 2022

Bucharest, Romania

- Architected Go-based backend services for digital commerce and enterprise workflow platforms using **Go 1.12–1.18**, REST APIs, PostgreSQL, Redis, and Kafka.
- Replaced several Java Spring Boot batch processors with lightweight **Go** services, reducing memory consumption by **33%** and improving job completion time by **27%**.
- Designed clean architecture modules with repository, service, and transport layers, making Go services easier to test, maintain, and extend across multiple delivery teams.
- Built Kafka 2.x consumers and producers for order lifecycle events, inventory sync, and notification pipelines, reducing missed event handling incidents by **29%**.
- Improved PostgreSQL 11–14 performance by optimizing transaction boundaries, adding composite indexes, and removing N+1 query patterns from high-traffic API endpoints.
- Solved a production timeout issue in checkout APIs by adding context deadlines, circuit breakers, Redis caching, and better upstream retry controls in Go services.
- Implemented Docker and Kubernetes deployment templates with Helm 3.x, helping teams standardize service configuration, probes, resource limits, and environment-specific values.
- Migrated legacy REST integrations into **gRPC** where internal service calls needed stronger contracts, lower payload overhead, and better long-term maintainability.
- Added structured logging, metrics, and distributed tracing across Go services, improving incident visibility and reducing debugging time by **32%** during critical releases.
- Reviewed pull requests, defined backend engineering standards, and supported developers with practical guidance on Go testing, API design, database migrations, and observability.

Tessenderlo Group — Senior Software Engineer

Remote/ Full time

Aug 2015 – Oct 2017

Brussel Metropolitan Area

- Developed backend services for logistics, planning, and manufacturing operations using **Go 1.7–1.10**, Java 8, PostgreSQL, RabbitMQ, and Docker.
- Introduced selected **Go** services for lightweight operational APIs where fast startup, simple deployment, and low runtime overhead were more effective than heavier application stacks.
- Improved inventory synchronization reliability by building message-driven processing with RabbitMQ 3.x and retry queues, reducing failed sync cases by **23%**.
- Migrated several older SOAP-style integrations into REST-based APIs, making warehouse and planning systems easier to integrate with modern internal applications.
- Solved recurring data mismatch issues by adding validation layers, reconciliation jobs, and stronger PostgreSQL constraints for shipment, inventory, and order status records.
- Improved reporting query performance by **26%** through index tuning, query refactoring, and background pre-aggregation for frequently accessed operational dashboards.
- Built Docker-based development environments for backend teams, reducing machine setup issues and improving local onboarding consistency across several internal projects.
- Collaborated with business analysts and operations users to convert manual logistics steps into automated backend workflows with clearer audit trails and fewer human errors.
- Supported production incidents by analyzing logs, database locks, queue backlogs, and API failures, then documenting fixes so repeated issues could be prevented.

Tessenderlo Group — Software Engineer

Hybrid/ Full time

Sep 2011 – Jul 2015

Brussel Metropolitan Area

- Built enterprise backend modules for manufacturing, finance, and logistics applications using Java 6–8, Spring Framework 3–4, SQL Server, PostgreSQL, and early REST APIs.
- Created service layers and database procedures for internal order processing systems, improving operational data consistency and reducing manual correction work by **21%**.
- Modernized legacy monolithic modules by separating business logic from UI-driven code paths, making later API extraction and service migration easier for engineering teams.
- Resolved a recurring production deadlock issue by reviewing transaction isolation, shortening long-running database operations, and adding safer retry handling around critical updates.
- Implemented REST endpoints for internal dashboards and reporting tools, helping business teams access shipment, planning, and finance data with fewer spreadsheet-based workflows.

- Improved nightly batch stability by adding validation checks, structured error logging, and restart-safe processing, reducing failed batch recovery effort by **25%**.
- Worked with QA teams to define regression scenarios for high-risk logistics workflows, improving release confidence around pricing, shipment status, and inventory calculations.
- Supported early containerization discussions and deployment automation by documenting service dependencies, configuration values, and environment differences across applications.

Tessengerlo Group — Software Developer

Hybrid/ Full time

Jun 2009 – Aug 2011

Brussel Metropolitan Area

- Developed internal business applications for manufacturing and supply chain teams using Java 5–6, PHP 5.x, SQL Server, Oracle, JavaScript, and server-rendered web interfaces.
- Fixed production data-entry defects by improving validation rules, database constraints, and error messages, reducing repeated user-submitted support cases by **18%**.
- Built reporting and export features for operational users, helping teams replace manual spreadsheet preparation with controlled application-generated reports.
- Supported migration of older database scripts into cleaner stored procedures and versioned deployment steps, reducing release mistakes during maintenance windows.
- Collaborated closely with senior engineers to understand legacy architecture, troubleshoot production issues, and gradually improve code quality without disrupting business-critical workflows.

EDUCATION

KU Leuven — Master's Degree, Computer Science

Sep 2003 – May 2009